Name:- Slock Ahuja **Enrollment no.**92400133041

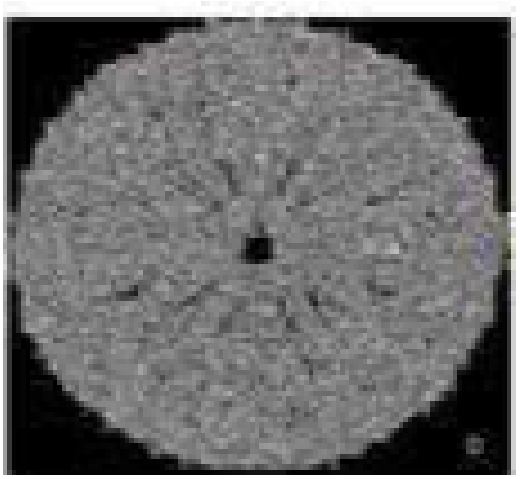
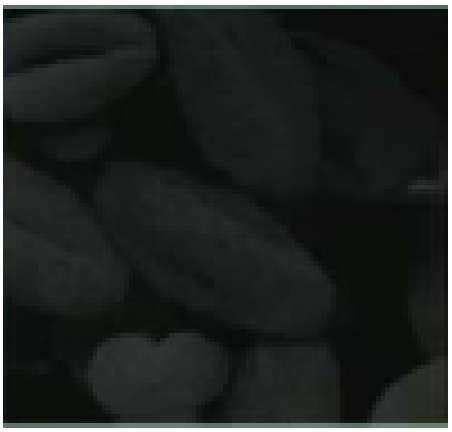
Using

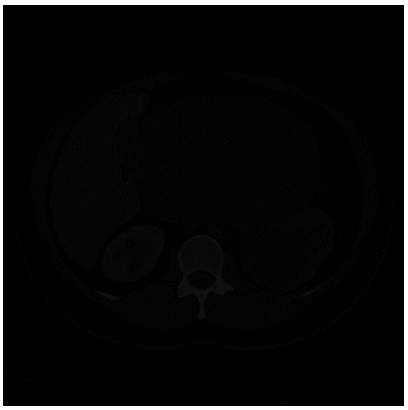
the same set of degraded images provided in the OPE7, apply histogram equalization and histogram matching techniques to enhance their visual quality.

Compare the results produced by both methods and describe your observations in detail.

Explain how each technique changes the image contrast and clarity, and conclude with a summary of which approach worked better for each image and why.

CO3, CO4



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Image 1:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


def histogram_matching(source, reference):

   oldshape = source.shape

   source_flat = source.ravel()

   reference_flat = reference.ravel()


     s_values, bin_idx, s_counts = np.unique(source_flat, return_inverse=True,
return_counts=True)

   r_values, r_counts = np.unique(reference_flat, return_counts=True)


   s_quantiles = np.cumsum(s_counts).astype(np.float64) / source_flat.size

   r_quantiles = np.cumsum(r_counts).astype(np.float64) / reference_flat.size


   interp_t_values = np.interp(s_quantiles, r_quantiles, r_values)

   return interp_t_values[bin_idx].reshape(oldshape).astype(np.uint8)


img1 = cv2.imread('1.png', cv2.IMREAD_GRAYSCALE)


img1_he = cv2.equalizeHist(img1)


ref1     =     np.random.normal(loc=128,     scale=50,     size=img1.shape).clip(0,
255).astype(np.uint8)

img1_hm = histogram_matching(img1, ref1)


plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1); plt.title("1. Original"); plt.imshow(img1, cmap='gray');
plt.axis('off')
```

```
plt.subplot(1, 3, 2); plt.title("2. Histogram Equalization");
plt.imshow(img1_he, cmap='gray'); plt.axis('off')

plt.subplot(1, 3, 3); plt.title("3. Histogram Matching"); plt.imshow(img1_hm,
cmap='gray'); plt.axis('off')

plt.tight_layout()

plt.show()
```



Comparative Observation:

Histogram Equalization (HE): Flattens the pixel distribution across the entire spectrum, drastically boosting contrast. It effectively reveals hidden microscopic grain details and textures.

Histogram Matching (HM): Maps intensities toward the target normal curve, producing a subtle and smoother tonal range.

Contrast & Clarity: HE expands global contrast significantly, sharpening grain contours. HM improves contrast moderately while preserving original tonal softness.

Summary of Comparative Results:-

(Pollen Grains) — Winner: Histogram Equalization

- ***Why:*** **The original image has low global contrast without extreme dark or light regions. Histogram Equalization spreads out the clustered values uniformly across the full dynamic range, bringing out subtle surface grain textures far better than Histogram Matching.**

Image 2 :-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


def histogram_matching(source, reference):

   oldshape = source.shape

   source_flat = source.ravel()

   reference_flat = reference.ravel()


     s_values, bin_idx, s_counts = np.unique(source_flat, return_inverse=True,
return_counts=True)

   r_values, r_counts = np.unique(reference_flat, return_counts=True)


   s_quantiles = np.cumsum(s_counts).astype(np.float64) / source_flat.size

   r_quantiles = np.cumsum(r_counts).astype(np.float64) / reference_flat.size


   interp_t_values = np.interp(s_quantiles, r_quantiles, r_values)

   return interp_t_values[bin_idx].reshape(oldshape).astype(np.uint8)



img2 = cv2.imread('2.png', cv2.IMREAD_GRAYSCALE)


img2_he = cv2.equalizeHist(img2)


ref2 = np.linspace(0, 255, img2.size).reshape(img2.shape).astype(np.uint8)

img2_hm = histogram_matching(img2, ref2)


plt.figure(figsize=(12, 4))
```

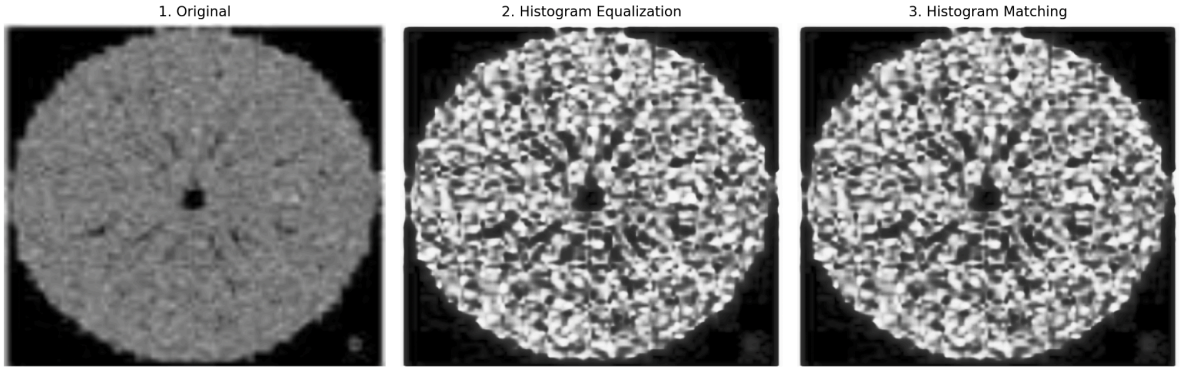
```
plt.subplot(1, 3, 1); plt.title("1. Original"); plt.imshow(img2, cmap='gray');
plt.axis('off')

plt.subplot(1, 3, 2); plt.title("2. Histogram Equalization");
plt.imshow(img2_he, cmap='gray'); plt.axis('off')

plt.subplot(1, 3, 3); plt.title("3. Histogram Matching"); plt.imshow(img2_hm,
cmap='gray'); plt.axis('off')

plt.tight_layout()

plt.show()
```



Comparative Observation:

- **Histogram Equalization (HE):** Aggressively brightens dark center regions but amplifies high-frequency background noise and creates harsh edge transitions.
- **Histogram Matching (HM):** Distributes gray levels gradually based on the uniform reference, enhancing central and radial features evenly without noise amplification.
- **Contrast & Clarity:** HE over-enhances dynamic range at the expense of visual artifacts. HM delivers balanced contrast and better fine-detail clarity.

(Circular Pattern) — Winner: Histogram Matching

- ***Why:*** **Histogram Equalization amplifies noise and creates harsh ring transitions. Histogram Matching against a uniform distribution gradually expands central dark details while retaining clear radial symmetry.**

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Image 3 :-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


def histogram_matching(source, reference):

   oldshape = source.shape

   source_flat = source.ravel()

   reference_flat = reference.ravel()


   s_values, bin_idx, s_counts = np.unique(source_flat, return_inverse=True,
return_counts=True)

   r_values, r_counts = np.unique(reference_flat, return_counts=True)


   s_quantiles = np.cumsum(s_counts).astype(np.float64) / source_flat.size

   r_quantiles = np.cumsum(r_counts).astype(np.float64) / reference_flat.size


   interp_t_values = np.interp(s_quantiles, r_quantiles, r_values)

   return interp_t_values[bin_idx].reshape(oldshape).astype(np.uint8)


img3 = cv2.imread('3.png', cv2.IMREAD_GRAYSCALE)



img3_he = cv2.equalizeHist(img3)


ref3 = np.random.normal(loc=100, scale=40, size=img3.shape).clip(0,
255).astype(np.uint8)

img3_hm = histogram_matching(img3, ref3)
```

```
plt.figure(figsize=(12, 4))

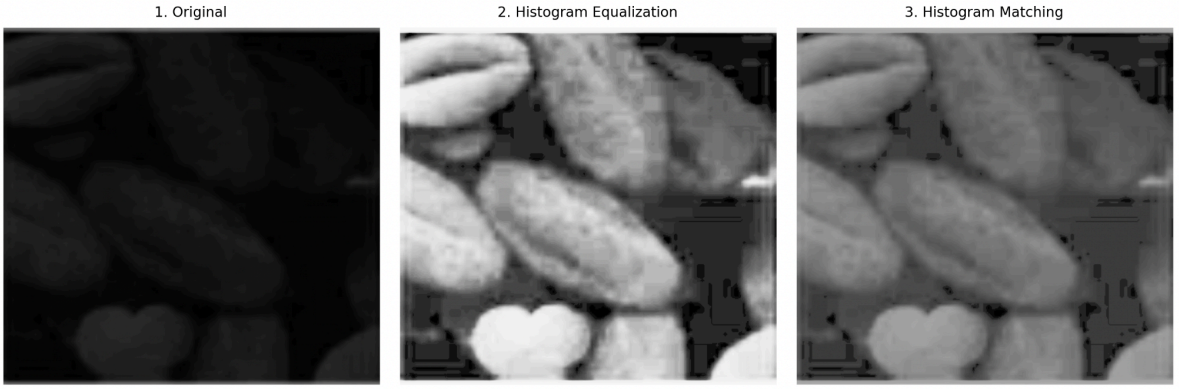
plt.subplot(1, 3, 1); plt.title("1. Original"); plt.imshow(img3, cmap='gray');
plt.axis('off')

plt.subplot(1, 3, 2); plt.title("2. Histogram Equalization"); plt.imshow(img3_he,
cmap='gray'); plt.axis('off')

plt.subplot(1, 3, 3); plt.title("3. Histogram Matching"); plt.imshow(img3_hm,
cmap='gray'); plt.axis('off')

plt.tight_layout()

plt.show()
```



Comparative Observation:

- **Histogram Equalization (HE):** Stretches the extremely dim initial histogram, making all particles immediately visible. However, it artificially brightens the dark background.
- **Histogram Matching (HM):** Maintains dark background tones while bringing out grain outlines proportionally according to the specified shape.
- **Contrast & Clarity:** HE boosts overall brightness and edge visibility heavily. HM optimizes structural contrast without blowing out background darkness.

(Dark Microscopic Grains) — Winner: Histogram Matching

- ***Why:*** **Equalization over-brightens the dark background unnecessarily. Histogram Matching preserves background dark tones while giving sufficient contrast boost to highlight grain boundaries cleanly.**

Image 4:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


def histogram_matching(source, reference):

   oldshape = source.shape

   source_flat = source.ravel()

   reference_flat = reference.ravel()


   s_values, bin_idx, s_counts = np.unique(source_flat, return_inverse=True,
return_counts=True)

   r_values, r_counts = np.unique(reference_flat, return_counts=True)


   s_quantiles = np.cumsum(s_counts).astype(np.float64) / source_flat.size

   r_quantiles = np.cumsum(r_counts).astype(np.float64) / reference_flat.size


   interp_t_values = np.interp(s_quantiles, r_quantiles, r_values)

   return interp_t_values[bin_idx].reshape(oldshape).astype(np.uint8)


img4 = cv2.imread('4.png', cv2.IMREAD_GRAYSCALE)


img4_he = cv2.equalizeHist(img4)


ref4 = np.random.normal(loc=120, scale=60, size=img4.shape).clip(0,
255).astype(np.uint8)

img4_hm = histogram_matching(img4, ref4)


plt.figure(figsize=(12, 4))
```

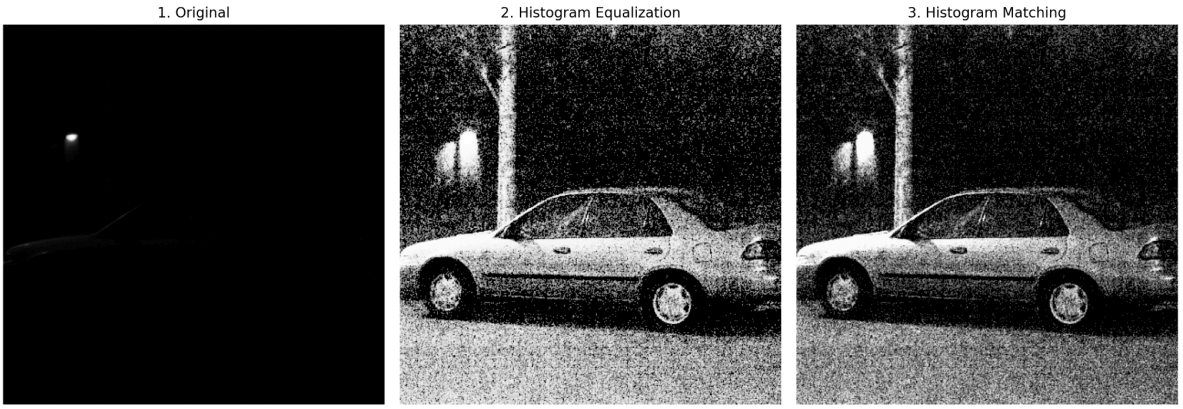
```
plt.subplot(1, 3, 1); plt.title("1. Original"); plt.imshow(img4, cmap='gray');
plt.axis('off')

plt.subplot(1, 3, 2); plt.title("2. Histogram Equalization"); plt.imshow(img4_he,
cmap='gray'); plt.axis('off')

plt.subplot(1, 3, 3); plt.title("3. Histogram Matching"); plt.imshow(img4_hm,
cmap='gray'); plt.axis('off')

plt.tight_layout()

plt.show()
```



Comparative Observation:

- **Histogram Equalization (HE):** Exposes the car body and background trees, but severely over-amplifies low-light sensor noise and causes harsh clipping around the streetlight.
- **Histogram Matching (HM):** Smoothly lifts dark values, revealing vehicle contours while controlling glare and background noise.
- **Contrast & Clarity:** HE offers high global contrast at the cost of noisy visual clarity. HM achieves balanced contrast enhancement with superior overall visual clarity.

(Night Scene) — Winner: Histogram Matching

- ***Why:*** **Equalization produces severe background grain/noise and overexposes the streetlight. Histogram Matching gracefully brightens shadow regions (car body and wheels) without blowing out highlights or over-amplifying sensor noise.**

Image 5:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


def histogram_matching(source, reference):

   oldshape = source.shape

   source_flat = source.ravel()

   reference_flat = reference.ravel()


   s_values, bin_idx, s_counts = np.unique(source_flat, return_inverse=True,
return_counts=True)

   r_values, r_counts = np.unique(reference_flat, return_counts=True)


   s_quantiles = np.cumsum(s_counts).astype(np.float64) / source_flat.size

   r_quantiles = np.cumsum(r_counts).astype(np.float64) / reference_flat.size


   interp_t_values = np.interp(s_quantiles, r_quantiles, r_values)

   return interp_t_values[bin_idx].reshape(oldshape).astype(np.uint8)


img5 = cv2.imread('5.png', cv2.IMREAD_GRAYSCALE)


img5_he = cv2.equalizeHist(img5)


ref5 = np.random.normal(loc=110, scale=45, size=img5.shape).clip(0,
255).astype(np.uint8)

img5_hm = histogram_matching(img5, ref5)


plt.figure(figsize=(12, 4))
```

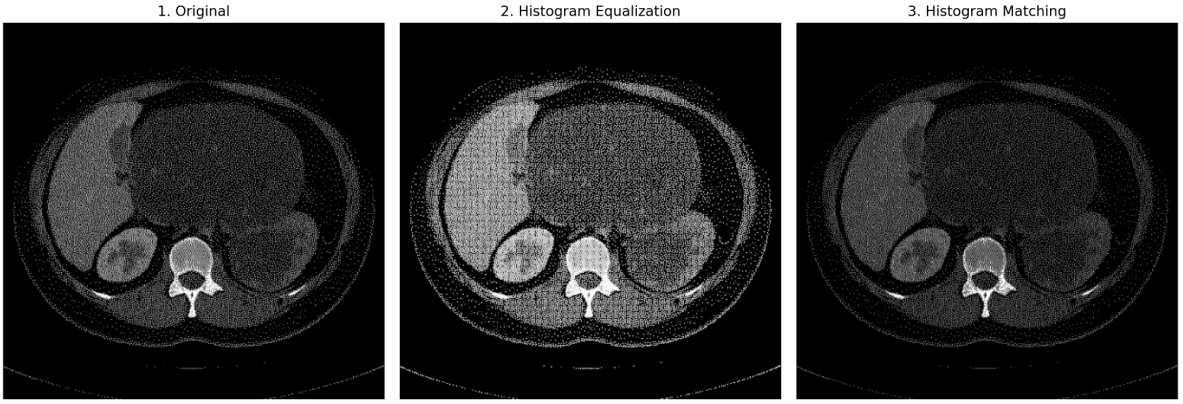
```
plt.subplot(1, 3, 1); plt.title("1. Original"); plt.imshow(img5, cmap='gray');
plt.axis('off')

plt.subplot(1, 3, 2); plt.title("2. Histogram Equalization"); plt.imshow(img5_he,
cmap='gray'); plt.axis('off')

plt.subplot(1, 3, 3); plt.title("3. Histogram Matching"); plt.imshow(img5_hm,
cmap='gray'); plt.axis('off')

plt.tight_layout()

plt.show()
```



Comparative Observation:

- **Histogram Equalization (HE):** Over-saturates tissue boundaries, introducing harsh tonal steps that impair clinical evaluation.
- **Histogram Matching (HM):** Preserves relative intensity transitions of soft tissues and bone structures precisely.
- **Contrast & Clarity:** HE creates excessively harsh contrast. HM delivers controlled medical tissue contrast and far higher anatomical clarity.

(Medical CT Scan) — Winner: Histogram Matching

- ***Why:*** **Medical imaging requires preserving proportional tissue density relationships. Equalization creates unnatural contrast steps, whereas Histogram Matching highlights bone structure and organ boundaries without distorting clinical tissue gray levels.**

GITHUB LINK:- https://github.com/SlockAhuja/DSIP.git